Customer Behaviour Analysis

Business Problem

How can a retail company understand customer purchasing behaviour to increase revenue, improve customer retention, and optimize marketing strategies?

EDA- (Exploratory Data Analysis using Python)

Importing the libraries

```
import pandas as pd
import matplotlib.pyplot as plt
```

Importing the data

```
df = pd.read_csv('customer_shopping_behavior.csv')
df.head(4)
```

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash	Weekly
521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card	Quarterly
3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash	Monthly
1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash	Annually

Initial Data Exploration and cleaning in jupyter notebook using pandas

df.head()
df.tail()
df.describe()
df.info()
df.dtypes
df.isnull().sum()
Checked unique values
Found inconsistencies between Item Purchased and Category
Created a mapping dictionary
Handling Missing Values
identified and removed duplicates
Converted column names

Data Analysis using SQL Server

1.which category contribute most to revenue.

category	highest_revenue
Electronics	486366.96
Accessories	194866.87
Clothing	183188.39
Footwear	111236.2
Outerwear	18524

Impact:

- Helps prioritize high-performing categories
- optimizes inventory planning
- improves marketing

2.Discounts may reduce profit without increasing sales.

discount_applied	total_amount	avg_purchase
Yes	491940.78	219.224947291951
No	502241.64	182.235719279022

Impact:

- Identify effectiveness of discounts
- Reduce unnecessary discounts costs
- Improve profit margins

3.understanding of revenue contribution by gender segments.

gender	total_revenue
Male	390964.53
Female	352060.29
Other	251157.6

Impact:

- Helps design targeted marketing campaign
- Improves customer segmentation strategy
- Enhance personalization efforts

4.Identify high spending customers who are also discount users.

customer_id	purchase_amount	discount_applied
4977	1499.76	Yes
4055	1493.56	Yes
4226	1472.22	Yes
4600	1460.51	Yes
4662	1459.59	Yes
4408	1456.31	Yes
5091	1451.07	Yes
4573	1441.51	Yes
4940	1439.75	Yes
4079	1438.53	Yes

Impact:

- Identifies premium discount sensitive customer
- Enable targeted discount campaigns
- Improves customer retention and revenue

5.product performance based on customer satisfaction. Top vs Bottom Products

item_purchased	avg_amount
Headphones	831.34
Laptop	805.5
Bag	735.77
Phone	734.48
Watch	685.23

item_purchased	avg_amount
Jacket	56.74
Sandals	57.5
Coat	57.61
Sweater	57.7
Handbag	57.89

Impact

- Promotes high performing products
- Improves low rated products
- Enhance customer experience

6.Unclear if faster shipping leads to higher spending.

shipping_type	avg_purchase	total_revenue
Express	358.12	429380.98
Standard	340.57	409020.45
2-Day Shipping	60.73	38080
Free Shipping	60.41	40777
Store Pickup	59.89	38931
Next Day Air	58.63	37993

Impact

- Helps optimize shipping pricing strategy
- Encourages premium shipping adoption
- Increases average order value

7.The effectiveness of subscription programs.

subscription_status	avg_purchase	total_revenue
Yes	270.58	427245.4
No	165.72	566937.02

Impact

- Validates subscription model performance
- Improves customer loyalty programs
- Increase customer lifetime value

8.Products which are overly dependent on discounts.

item_purchased	total_no_of_item	discount	discount_percent
Laptop	143	74	51.750000000000
Phone	171	88	51.460000000000
Headphones	166	85	51.200000000000
Hat	154	77	50.000000000000
Watch	157	78	49.680000000000

Impact

- Identifies discount driven products
- Helps optimize pricing strategy
- Reduce profit margin loss

9.Lack of customer segmentation leads to generic strategies.

customer_segment	customer_count
New customer	561
Returning customer	1354
Loyal customer	3085

Impact

- Enables personalized marketing
- Improves retention strategies
- Increase conversion rates

10.Top performing products within categories.

category	item_purchased	most_purchased	mk
Accessories	Jewelry	171	1
Accessories	Bag	164	2
Accessories	Belt	161	3
Accessories	Sunglasses	161	3
Clothing	Shirt	315	1
Clothing	Blouse	171	2
Clothing	Pants	171	2
Clothing	Dress	166	3

Impact

- Improves product placement & recommendations.
- Helps inventory optimization
- Increase sales through best sellers

11.Relationship between loyalty and subscription

customer_type	subscription_status	customer_count
Repeat Buyers	No	2763
Normal Buyers	Yes	389
Normal Buyers	No	658
Repeat Buyers	Yes	1190

Impact

- Improves subscription targeting
- Increase conversion to paid programs
- Increase customer retention strategy

12.which age group contribution most to revenue

age_group	total_revenue
51+	399345.32
35-50	266706.7
25-35	177421.93
18-25	150708.48

Impact

- Enables age based targeting
- Enables marketing efficiency

Conclusion

The customer behaviour analysis shows that customer purchasing decisions are influenced by product category, discounts, subscription status, and purchase frequency. Loyal and repeat customers contribute a major share of the revenue, while some product categories perform significantly better than others. The analysis also highlights the importance of customer segmentation and personalized marketing strategies for improving sales and retention.

Recommendations

- Focus more on high-performing product categories to maximize revenue growth.

- Introduce loyalty and reward programs to retain repeat customers.
- Use targeted discounts instead of offering discounts to all customers.
- Promote subscription programs to increase customer engagement and recurring sales.
- Improve personalized marketing based on customer purchase behaviour and segmentation.
- Maintain proper inventory planning for high-demand products and seasonal sales.
- Enhance customer experience through better shipping services and faster delivery.

Power BI Dashboard